PHY202 Final Exam on chapter(s): 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28

1. $F = 1.09\,\text{N} \because F = Eq, \Delta V = Ed \because$ to $1\,e^-$ the gun supplies $E = 70.\,\text{keV}$, its $\Delta V = 70.\,\text{kV}$.
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cp1e_ch19_ex5)

2. $Q = 3.89 \times 10^{-9}\,\text{C} \because C = \dfrac{Q}{V} = \epsilon_0 \dfrac{A}{d}$
(Chapter 19, Section (5) Capacitors and Dielectrics - cp1e_ch19_ex8)

3. $\Delta V_{max} = 3.90 \times 10^{9}\,\text{V} \because \Delta V = Ed$
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cp1e_ch19_ex4)

4. $\varepsilon = 14.6\,\text{V} \because \varepsilon = NAB\omega \sin(\omega t), \sin(\omega t) = 1$
(Chapter 23, Section (5) Electric Generators - cp1e_ch23_ex4)

5. $\beta = 95.1\,\text{dB} \because v_w = \left(331\,\dfrac{\text{m}}{\text{s}}\right)\sqrt{\dfrac{T}{273\,\text{K}}}, I = \dfrac{(\Delta p)^2}{2\rho v_w}, \beta(\text{dB}) = 10\log_{10}\left(\dfrac{I}{I_0}\right), I_0 = 1 \times 10^{-12}\,\dfrac{\text{W}}{\text{m}^2}$
(Chapter 17, Section (3) Sound Intensity and Sound Level - cp1e_ch17_ex2)

6. $D = 7.31 \times 10^{-5}\,\text{m} \because R = \dfrac{\rho L}{A}$
(Chapter 20, Section (3) Resistance and Resistivity - cp1e_ch20_ex5)

7. $\lambda = 2.31 \times 10^{-6}\,\text{m} \because d\sin\theta = m\lambda$
(Chapter 27, Section (3) Young's Double-Slit Experiment - cp1e_ch27_ex1)

8. $L = 0.4733\,\text{m} \because f_n = n\dfrac{v_w}{4L}, v_w = \left(331\,\dfrac{\text{m}}{\text{s}}\right)\sqrt{\dfrac{T}{273\,\text{K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5a)

9. $m = -0.387 \because \dfrac{1}{d_0} + \dfrac{1}{d_i} = \dfrac{1}{f}, \dfrac{h_i}{h_0} = -\dfrac{d_i}{d_0} = m$
(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex6b)

10. $x_3 = 0.866\,\text{m} \because \Sigma \vec{F}_x = m\vec{a}_x, F = k\dfrac{|q_1 q_2|}{r^2}$
(Chapter 18, Section (3) Coulomb's Law - PSE_Serway_4e_ch23_ex3)

11. $\theta_c = 38.7^\circ \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
(Chapter 25, Section (4) Total Internal Reflection - cp1e_ch25_ex4)

12. $f = 4.37 \times 10^{6}\,\text{Hz} \because T = \dfrac{1}{f}$
(Chapter 16, Section (2) Period and Frequency in Oscillations - cp1e_ch16_ex3)

13. $I_p = 3.57\,\text{A} \because \dfrac{V_s}{V_p} = \dfrac{N_s}{N_p} = \dfrac{I_p}{I_s}$
(Chapter 23, Section (7) Transformers - cp1e_ch23_ex6)

14. $f = 4.65 \times 10^{8}\,\mathrm{Hz} \because \lambda_{obs} = \lambda_s \sqrt{\dfrac{1+\dfrac{u}{c}}{1-\dfrac{u}{c}}}, c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}} = \dfrac{E}{B} = f\lambda$

(Chapter 28, Section (4) Relativistic Addition of Velocities - cp1e_ch28_ex5b)

15. $B = 0.209\,\mathrm{T} \because r = \dfrac{mv}{qB}, m_p = 1.67 \times 10^{-27}\,kg, q_p = 1.60 \times 10^{-19}\,C$

(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - tkd1)

16. $E_{ind} = 1.08\,\mathrm{J}\ \because E_{ind} = \frac{1}{2} L I^2$

(Chapter 23, Section (9) Inductance - cp1e_ch23_ex8)

17. $a = 0.895\,\%\ \because Z = \rho v\,, a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$

(Chapter 17, Section (7) Ultrasound - cp1e_ch17_ex7)

18. $R = 5.82\,\Omega\ \because V = IR$

(Chapter 20, Section (2) Ohm's Law - cp1e_ch20_ex4)

19. $P = 7.69\,\mathrm{D}\ \because P = \frac{1}{f}$

(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex5)

20. $B_{max} = 1.30 \times 10^{-6}\,\mathrm{T}\ \because c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E}{B} = f\lambda$

(Chapter 24, Section (2) Production of Elecromagnetic Waves - cp1e_ch24_ex1)

21. $I_s = 2.61 \times 10^{-3}\,\mathrm{A}\ \because \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$

(Chapter 23, Section (7) Transformers - cp1e_ch23_ex5)

22. $L = 3.58\,\mathrm{cm}\ \because T_{pend} = 2\pi \sqrt{\frac{L}{g}}$

(Chapter 16, Section (4) Simple Pendulum - cp1e_ch16_p22)

23. $C_T = 2.20 \times 10^{-9}\,\mathrm{F}\ \because \frac{1}{C_s} = \frac{1}{C_1} + \ldots + \frac{1}{C_n}$

(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ch19_ex9)

24. $Q = 1.35 \times 10^{3}\,\mathrm{C}\ \because I = \frac{\Delta Q}{\Delta t}$

(Chapter 20, Section (1) Current - tkd0)

25. $f = 1.12 \times 10^{3}\,\mathrm{Hz}\ \because f_n = n \frac{v_w}{4L}, v_w = \left(331 \frac{\mathrm{m}}{\mathrm{s}}\right) \sqrt{\frac{T}{273\,\mathrm{K}}}$

(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5b)

26. $F = 1.35 \times 10^{-12}\,\mathrm{N}\ \because F = qvb \sin\theta\,, \theta = 90^\circ$

(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ch22_ex1)

27. $I = 1.18 \times 10^{3}\,\mathrm{A}\ \because I = \frac{\Delta Q}{\Delta t}$

(Chapter 20, Section (1) Current - cp1e_ch20_ex1)

28. $k = 5.88 \times 10^{4} \frac{\mathrm{N}}{\mathrm{m}}\ \because \vec{F} = -k \Delta\vec{x} \cdot k = -\frac{\vec{F}}{\Delta\vec{x}}$

(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex1)

29. $P = 52.6\,\text{D} \because \dfrac{1}{d_0} + \dfrac{1}{d_i} = \dfrac{1}{f}, P = \dfrac{1}{f}$
(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex2a)

30. $F = 3.89 \times 10^{-6}\,\text{N} \because F = qvb\sin\theta \therefore F_{max} = qvb$
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ch22_p7)

31. $I = 4.96\,\text{A} \because \dfrac{1}{R_p} = \dfrac{1}{R_1} + ... + \dfrac{1}{R_n}, V = IR$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex1)

32. $I_{RMS} = 75.0\,\text{A} \because P_{ave} = I_{RMS} V_{RMS}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex10)

33. $\varepsilon = 1.41 \times 10^{-4}\,\text{V} \because \varepsilon = Blv$
(Chapter 22, Section (6) The Hall Effect - cp1e_ch22_p22)

34. $I = 5.98 \times 10^{-12}\,\text{A} \because I(t) = I_0 e^{-t/\tau}, \tau = \dfrac{L}{R}$
(Chapter 23, Section (10) RL Circuits - cp1e_ch23_ex9b)

35. $n_2 = 1.16 \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
(Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex2)

36. $P = -4.42\,\text{D} \because \dfrac{1}{d_0} + \dfrac{1}{d_i} = \dfrac{1}{f}, P = \dfrac{1}{f}$
(Chapter 26, Section (2) Vision Correction - cp1e_ch26_ex3)

37. $\varepsilon = 9.52 \times 10^{3}\,\text{V} \because \varepsilon = Blv$
(Chapter 23, Section (3) Motional EMF - cp1e_ch23_ex2)

38. $I = 0.543\,\text{A} \because$ Kirchhoff's rules
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ1)

39. $E = 3.16 \times 10^{3}\,\text{J} \because P = IV = \dfrac{E}{t}, V = \text{emf} - Ir$
(Chapter 21, Section (2) Electromotive Force: Terminal Voltage - cp1e_ch21_ex4)

40. $t = 7.75 \times 10^{-8}\,\text{m} \because 2t = \left(m + \dfrac{1}{2}\right)\lambda_n, n_{\text{air}} < n_{\text{oil}} > n_{\text{water}}$
(Chapter 27, Section (7) Thin-Film Interference - cp1e_ch27_p72)

41. $P = 1.38\,\text{W} \because R_s = R_1 + R_2 + R_3, V = IR, P = IV$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex2)

42. $\dfrac{F}{l} = 140.\dfrac{\text{N}}{\text{m}} \because F = IlB\sin\theta, \theta = 90^\circ$
(Chapter 22, Section (7) Magnetic Force on a Current-Carrying Conductor - cp1e_ch22_ex4)

43. $\dot{N}_e = 4.19 \times 10^{20}$ electrons per second $\because$

$$\Delta V = \frac{\Delta PE}{q}, W_{c.f.} = \Delta PE \therefore W = -q\,\Delta V \therefore \frac{q}{t} = \frac{-W}{t\,\Delta V} \therefore \dot{N}_e = \frac{q}{q_e t} = \frac{-W}{q_e t\,\Delta V}$$

(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ch19_ex2)

44. $f = 0.671\,\mathrm{Hz} \because T_{SHO} = 2\pi\sqrt{\frac{m}{k}}, f = \frac{1}{T}$

(Chapter 16, Section (3) Simple Harmonic Motion - cp1e_ch16_ex4)

45. $D = 1.40 \times 10^{-6}\,\mathrm{m} \because D\sin\theta = m\lambda$

(Chapter 27, Section (5) Single-Slit Diffraction - cp1e_ch27_ex4)

46. $P = 52.6\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$

(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex2b)

47. $\tau = 0.0157\,\mathrm{s} \because \tau = \frac{L}{R}$

(Chapter 23, Section (10) RL Circuits - cp1e_ch23_ex9)

48. $\mathrm{emf} = 95.7\,\mathrm{V} \because \mathrm{emf} = -N\frac{\Delta\Phi}{\Delta t}$

(Chapter 23, Section (2) Faraday's Law of Induction: Lenz's Law - cp1e_ch23_p8)

49. $L = 5.34 \times 10^{-5}\,\mathrm{H} \because L = N\frac{\Delta\Phi}{\Delta t} = \frac{\mu_0 N^2 A}{l}$

(Chapter 23, Section (9) Inductance - cp1e_ch23_ex7)

50. $E = 1.17 \times 10^{14}\,\mathrm{J} \because E_0 = mc^2$

(Chapter 28, Section (6) Relativistic Energy - cp1e_ch28_ex6)

51. $E = 4.68 \times 10^{4}\,\mathrm{N/C} \because E = k\frac{|Q|}{r^2}$

(Chapter 18, Section (4) Electric Field - cp1e_ch18_ex2)

52. $v_f = 4.78 \times 10^{5}\,\frac{\mathrm{m}}{\mathrm{s}} \because \Delta V = \frac{\Delta PE}{q}, W_{c.f.} = -\Delta PE \therefore W = -q\,\Delta V, W_{net} = \Delta KE \therefore \Delta KE = -q\,\Delta V$

(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ch19_ex3)

53. $m = -1.67 \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$

(Chapter 26, Section (6) Image Formation by Lenses - cp1e_ch25_ex7)

54. $B = 1.05 \times 10^{-4}\,\mathrm{T} \because r = \frac{mv}{qB}, m_e = 9.11 \times 10^{-31}\,kg, q_e = 1.60 \times 10^{-19}\,C$

(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ch22_p13)

55. $I = 2.03\,\mathrm{A} \because$ Kirchhoff's rules

(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)

56. $\theta = 0.0411\,\text{arcseconds} \because \theta = 1.22\dfrac{\lambda}{D}$

(Chapter 27, Section (6) Limits of Resolution - cp1e_ch27_ex5)

57. $\theta_2 = 3.15^{\circ} \because n_1 \sin\theta_1 = n_2 \sin\theta_2$

(Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex3)

58. $\%\Delta P = -23.9\% \because \rho(T) = \rho_0(1+\alpha\,\Delta T), R = \dfrac{\rho L}{A}, P = \dfrac{V^2}{R}$

(Chapter 20, Section (4) Electric Power and Energy - tkd3)

59. $Q = 3.30 \times 10^{-6}\,\text{C} \because V = \dfrac{kQ}{r}$

(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cp1e_ch19_ex7)

60. $\Delta V = 17.5\,\text{V} \because V = \dfrac{kQ}{r}$

(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cp1e_ch19_ex6)

61. $v = 2.31 \times 10^{8}\,\dfrac{\text{m}}{\text{s}} \because n = \dfrac{c}{v}$

(Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex1)

62. $\Delta y = 4.63 \times 10^{-12}\,\text{m} \because F = qvb\sin\theta, F_{net} = ma, \vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \dfrac{1}{2}\vec{a}t^2$

(Chapter 22, Section (11) Applications of Magnetism - cp1e_ch22_p82)

63. $C_T = 1.06 \times 10^{-8}\,\text{F} \because \dfrac{1}{C_s} = \dfrac{1}{C_1} + \ldots + \dfrac{1}{C_n}, C_p = C_1 + \ldots + C_n$

(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ch19_ex10)

64. $x = 4.25 \times 10^{3}\,\text{m} \because \theta = 1.22\dfrac{\lambda}{D}, \Delta\theta = \dfrac{\Delta s}{r}, \Delta s \approx 0.70\,\text{m}$ for small angles

(Chapter 27, Section (6) Limits of Resolution - cp1e_ch27_p66)

65. $W_{ratio} = 171.$ times more energy $\because \Delta V = \dfrac{\Delta PE}{q}, W = -\Delta PE$ for a conservative force, so $W = -q\,\Delta V$

(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cp1e_ch19_ex1)

66. $d = 1.03 \times 10^{-6}\,\text{m} \because P = \dfrac{V^2}{R}, R = \dfrac{\rho L}{A}$ and $A = \pi r^2$

(Chapter 20, Section (4) Electric Power and Energy - tkd2)

67. $E_0 = 1.67 \times 10^{3}\,\dfrac{\text{V}}{\text{m}} \because I_{ave} = \dfrac{c\,\varepsilon_0 E_0^2}{2} = \dfrac{c\,B_0^2}{2\mu_0} = \dfrac{E_0 B_0}{2\mu_0}$

(Chapter 24, Section (4) Energy in Electromagnetic Waves - cp1e_ch24_ex4)

68. $v_{max} = 0.363\,\dfrac{\text{m}}{\text{s}} \because v(t) = \sqrt{\dfrac{k}{m}}\,X \sin\dfrac{2\pi t}{T}$ and the max of sine's range is 1.

(Chapter 16, Section (5) Energy and the Simple Harmonic Oscillator - cp1e_ch16_ex6)

69. $\varepsilon = 7.35\,\text{V} \because \varepsilon = -N\dfrac{\Delta\Phi}{\Delta t}$
(Chapter 23, Section (5) Electric Generators - cp1e_ch23_ex3)

70. $d_i = 0.186\,\text{m} \because \dfrac{1}{d_0} + \dfrac{1}{d_i} = \dfrac{1}{f}$
(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex6a)

71. $h_i = -1.99 \times 10^{-5}\,\text{m} \because \dfrac{h_i}{h_0} = -\dfrac{d_i}{d_0} = m$
(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex1)

72. $q = 2.78 \times 10^{-8}\,\text{C} \because \Sigma\vec{F}_x = m\vec{a}_x, F = k\dfrac{|q_1 q_2|}{r^2}$
(Chapter 18, Section (3) Coulomb's Law - PSE_Serway_4e_ch23_ex5)

73. $\Delta t = 1.34 \times 10^{-6}\,\text{s} \because \gamma = \left(\sqrt{1 - \dfrac{v^2}{c^2}}\right)^{-1}, \Delta t = \gamma\,\Delta t_0$
(Chapter 28, Section (2) Simultaneity and Time Dilation - cp1e_ch28_ex1)

74. $r = 0.491\,\text{m} \because r = \dfrac{mv}{qB}, m_p = 1.67 \times 10^{-27}\,kg, q_p = 1.60 \times 10^{-19}\,C$
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ch22_ex2)

75. $\theta = 71.6^\circ \because I = I_0 \cos^2\theta$
(Chapter 27, Section (8) Polarization - cp1e_ch27_p87)

76. $F = 2.99 \times 10^{-5}\,\text{N} \because F = Eq$
(Chapter 18, Section (4) Electric Field - cp1e_ch18_ex3)

77. $R_{eq} = 2.12\,\Omega \because R_s = R_1 + R_2 + R_3, \dfrac{1}{R_p} = \dfrac{1}{R_1} + \ldots + \dfrac{1}{R_n}$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex3)

78. $P = 10.6\,\text{D} \because \dfrac{1}{d_0} + \dfrac{1}{d_i} = \dfrac{1}{f}, P = \dfrac{1}{f}$
(Chapter 26, Section (2) Vision Correction - cp1e_ch26_ex4)

79. $L = 1.82\,\text{ly} \because L = L_0\gamma^{-1}$
(Chapter 28, Section (3) Length Contraction - cp1e_ch28_ex2)

80. $f = 560.5\,\text{Hz} \because f_{obs} = f_s\left(\dfrac{v_w}{v_w \pm v_s}\right)$ (review the Doppler effect)
(Chapter 17, Section (4) Doppler Effect and Sonic Booms - cp1e_ch17_ex4)

81. $I = 12.0\,\text{A} \because B = \dfrac{\mu_0 I}{2\pi r}$
(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ch22_ex6)

82. $d = 0.787\,\mathrm{m} \because \Delta E_t = W_{nc} = -f_k d, f_k = \mu_k N = \mu_k mg, PE = \frac{1}{2} k x^2$
(Chapter 16, Section (7) Damped Harmonic Motion - cp1e_ch16_ex7)

83. $\varepsilon = 1.63 \times 10^{-4}\,\mathrm{V} \because \varepsilon = B l v$
(Chapter 22, Section (6) The Hall Effect - cp1e_ch22_ex3)

84. $F = 4.64 \times 10^{-6}\,\mathrm{N} \because F = k \frac{|q_1 q_2|}{r^2}$
(Chapter 18, Section (3) Coulomb's Law - cp1e_ch18_ex1)

85. $F = 25.1\,\mathrm{N} \because \frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r}$
(Chapter 22, Section (10) Magnetic Force betewen Two Parallel Conductors - cp1e_ch22_p50)

86. $PE = 0.510\,\mathrm{J} \because PE = \frac{1}{2} k x^2$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex2)

87. $E = 3.36 \times 10^7 \frac{\mathrm{N}}{\mathrm{C}} \because E = k \frac{|Q|}{r^2}, E^2 = E_x{}^2 + E_y{}^2$
(Chapter 18, Section (5) Electric Field Lines - cp1e_ch18_ex4)

88. $d_{min} = 2.25 \times 10^{-3}\,\mathrm{m} \because R = \frac{\rho L}{A}, P_{\int ¿ = I^2 R ¿}$
(Chapter 20, Section (4) Electric Power and Energy - tkd4)

89. $I = 21.0\,\mathrm{A} \because B = \frac{\mu_0 I}{2R}$
(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ch22_ex7)

90. $E = 8.70 \times 10^3\,\mathrm{J} \because P = IV = \frac{E}{t}$
(Chapter 20, Section (4) Electric Power and Energy - tkd1)

91. $\theta = 24.9^{\circ} \because d \sin\theta = m\lambda$
(Chapter 27, Section (4) Multiple-Slit Diffraction - cp1e_ch27_ex3)

92. $d = 823.\,\mathrm{m} \because \vec{v}_{avg} = \frac{\Delta \vec{x}}{\Delta t}$
(Chapter 17, Section (2) Speed of Sound, Frequency, Wavelength - cp1e_ch17_p8)

93. $\lambda_{obs} = 0.696\,\mathrm{m} \because \lambda_{obs} = \lambda_s \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}}$
(Chapter 28, Section (4) Relativistic Addition of Velocities - cp1e_ch28_ex5)

94. $C = 1.95 \times 10^{-6}\,\mathrm{F} \because E_{cap} = \frac{C V^2}{2}$
(Chapter 19, Section (7) Energy Stored in Capacitors - cp1e_ch19_ex11)

95. $F = 1.35 \times 10^{-12}\,\mathrm{N} \because F = Eq$
(Chapter 18, Section (8) Applications of Electrostatics - cp1e_ch18_ex5)

96. $P = 57.6\,\mathrm{W} \because P = \dfrac{V^2}{R}$
(Chapter 20, Section (4) Electric Power and Energy - cp1e_ch20_ex7)

97. $\tau = 18.9\,\mathrm{N} \cdot \mathrm{m} \because \tau = NIAB\sin\theta \therefore \tau_{max} = NIAB$
(Chapter 22, Section (8) Torque on a Current Loop: Motors and Meters - cp1e_ch22_ex5)

98. $V_0 = 255.\,\mathrm{V} \because V_{RMS} = \dfrac{V_0}{\sqrt{2}}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex9)